

**Project Design Phase**  
**Problem – Solution Fit Template**

|               |                                |
|---------------|--------------------------------|
| Date          | 31 July 2026                   |
| Team ID       | Panchumarthi Durga Kosali      |
| Project Name  | Metro Ticket Generation System |
| Maximum Marks | 2 Marks                        |

**Problem – Solution Fit Template:**

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

**Purpose:**

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Benefits: Generating metro tickets instantly ,reducing waiting time for passengers even reduces the manual work and automates the date and destination, source, fare . Ensures secure handling of ticket information generates reports and analyzing.

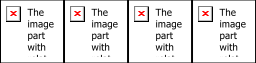
Key Features: User registration/login, route selection, fare calculation, QR ticket generation, online payment, booking history, ticket cancellation .

Proposed Solution: A ServiceNow-based Metro Ticket Generation System that allows users to generate digital tickets, select source and destination stations, make secure online payments, receive QR-code tickets, and view ticket history.

Current Challenges: Manual ticket purchase, waiting time, cash dependency, ticket loss, and slow transaction processing.

Target Users: Daily commuters, occasional travelers, tourists, and metro station staff.

Problem Statement: Daily metro commuters face long queues at ticket counters, delays in purchasing tickets, and inconvenience during peak hours.



Project Design Phase

Problem – Solution Fit Template

Date

15 February 2025

| Team ID                                                                                                                                                                                                                                                                                                                                                                                                  | MTG-01                                                                                                                                                                                                                                                                                                                                                | Project Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Metro Ticket Generation                                                                                                                                                                                                                                                     | Maximum Marks                                                                                                                                                                                                                                                                                                                                               | 2 Marks |
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| <div><div>Problem – Solution Fit Template:</div><div>The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.</div></div>                            |                                                                                                                                                                                                                                                                                                                                                       | <div><div>Purpose:</div><div><ul style="list-style-type: none"><li>Solve complex problems in a way that fits the state of your customers.</li><li>Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.</li><li>Sharpen your communication and marketing strategy with the right triggers and messaging.</li><li>Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.</li><li>Understand the existing situation in order to improve it for your target group.</li></ul></div></div> <div></div> |                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                             |         |
| <div><div>1. THE PROBLEM</div><div>Passengers face difficulties in purchasing metro tickets due to:<ul style="list-style-type: none"><li>Long queues at ticket counters</li><li>Lack of real-time seat/route information</li><li>Cash dependency and no digital receipt</li><li>Difficulty in calculating fare for different routes</li><li>Risk of losing paper tickets</li></ul></div></div>           | <div><div>2. TARGET USERS</div><div><ul style="list-style-type: none"><li>Daily Metro Commuters</li><li>Tourists / Visitors</li><li>Students</li><li>Senior Citizens</li><li>Working Professionals</li></ul></div></div>                                                                                                                              | <div><div>3. CURRENT BEHAVIOR (AS-IS)</div><div><ul style="list-style-type: none"><li>Go to ticket counter or kiosk</li><li>Ask for route and fare</li><li>Wait in queue</li><li>Pay using cash or card</li><li>Receive paper token/ticket</li><li>Validate ticket at entry gate</li><li>Exit after journey</li></ul></div></div>                                                                                                                                                                                                                                                                                                                                     | <div><div>4. EXISTING SOLUTION (ALTERNATIVES)</div><div><ul style="list-style-type: none"><li>Manual Ticket Counters</li><li>Token Vending Machines</li><li>Smart Cards (Limited Functionality)</li><li>Third-party Mobile Apps (Limited integration)</li></ul></div></div> | <div><div>5. PROPOSED SOLUTION (TO-BE)</div><div>A digital Metro Ticket Generation System that allows passengers to search routes, calculate fare, make payments and generate QR-based tickets digitally for seamless metro travel.</div></div>                                                                                                             |         |
| <div><div>6. KEY FEATURES OF THE SOLUTION</div><div><ul style="list-style-type: none"><li>User Registration / Login</li><li>Route Search &amp; Fare Calculation</li><li>Multiple Payment Options (UPI, Card, Wallet)</li><li>QR Code Ticket Generation</li><li>Digital Ticket Storage</li><li>Ticket History</li><li>Real-time Notifications</li><li>Refund / Cancellation Support</li></ul></div></div> | <div><div>7. VALUE PROPOSITION</div><div>Our solution provides:<ul style="list-style-type: none"><li>Fast and convenient ticketing</li><li>Reduced waiting time</li><li>Secure and cashless transactions</li><li>Digital tickets – no risk of loss</li><li>Real-time information and updates</li><li>Enhanced travel experience</li></ul></div></div> | <div><div>8. CUSTOMER BENEFITS</div><div><ul style="list-style-type: none"><li>Saves Time</li><li>Easy and Accessible Anytime</li><li>Cashless &amp; Secure</li><li>No need to carry physical tickets</li><li>Better travel planning</li><li>Reliable and user-friendly experience</li></ul></div></div>                                                                                                                                                                                                                                                                                                                                                              | <div><div>9. EXPECTED OUTCOME</div><div>Increased passenger satisfaction and adoption of digital ticketing, reduced crowd at counters, efficient metro operations and contribution towards smart city digital transformation.</div><div></div></div>                        |                                                                                                                                                                                                                                                                                                                                                             |         |
| <div><div>10. SOLUTION WORKFLOW</div><div><div>1. Login / Register</div><div>→</div><div><div>2. Select Source &amp; Destination</div><div>→</div><div><div>3. Calculate Fare</div><div>→</div><div><div>4. Make Payment</div><div>→</div><div><div>5. QR Ticket Generated</div><div>→</div><div><div>6. Travel &amp; Validate Ticket</div></div></div></div></div></div></div></div>                    |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                             | <div><div>REFERENCES:</div><div><div>1. <a href="https://www.ideahackers.network/problem-solution-fit-canvas/">https://www.ideahackers.network/problem-solution-fit-canvas/</a></div><div>2. <a href="https://medium.com/@epicantus/problem-solution-fit-canvas">https://medium.com/@epicantus/problem-solution-fit-canvas</a></div></div><div></div></div> |         |

